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Specifications of XMP Camera Tags – Pix4D Software

The XMP namespace Camera is defined by Pix4D, URI: http://pix4d.com/camera/1.0

(XMP) Camera
(XMP) Radiometric correction
(XMP) Radiometric calibration target
(XMP) Position and orientation
(XMP) Sun irradiance position and orientation
(XMP) Bit-depth reduction
(XMP) Miscellaneous

> **Note:** For projects that require radiometric calculations (e.g. agriculture applications), refer to the camera XMP section as well as this page which gives the specifications for the different corrections to be applied: Exif/Xmp tags for radiometric correction.

(XMP) Camera

Key	Value example	Type	Meaning
Xmp.Camera.ModelType	"fisheye"	XmpText	Camera model type: one of "perspective" or "fisheye".
Xmp.Camera.PrincipalPoint	2.42299, 1.850154	XmpText	c_x, c_y in millimeters separated by comma.
Xmp.Camera.PerspectiveFocalLength	4.569165	XmpText	For perspective camera: focal length in millimeters.
Xmp.Camera.PerspectiveDistortion	0.201690322, -0.358534038, 0.256185599, 0.004208643, 0.000268361	XmpText	For perspective camera: R1, R2, R3, T1, T2 separated by a comma: How are the Internal and External Camera Parameters defined?.
Xmp.Camera.FisheyeAffineMatrix	1583.319083002,0,0,1583.319083002	Ascii	For fisheye camera: Affine matrix C, D, E, F separated by comma: How are the Internal and External Camera Parameters defined?.
Xmp.Camera.FisheyeAffineSymmetric	1	Boolean	For fisheye camera: If symmetric affine

			matrix should be used (default false).
Xmp.Camera.FisheyePolynomial	0, 1, -0.025178575, 0.167834214	Ascii	For fisheye camera: Polynomial separated by comma: How are the Internal and External Camera Parameters defined?.
Xmp.Camera.RigName	Airinov multiSPEC4C 1.1	XmpText	Name of the rig model this image belongs to.
Xmp.Camera.RigCameraIndex	0	Short	Corresponding camera index within the rig model.
Xmp.Camera.RigRelatives	0.3836, -1.27665, -0.1156	XmpText	Rotation angles with respect to the reference camera. For angle convention in Pix4D: Yaw, Pitch, Roll and Omega, Phi, Kappa angles.
Xmp.Camera.CaptureUUID	B9A2F3676304A83F92261B5245124F28	XmpText	Unique identifier for one capture within a flight. This allows us to find that capture across different rig cameras.
Xmp.Camera.FlightUUID	7F8DC95FFCADA339805BD53338EC888 E	XmpText	Unique identifier for one flight
Xmp.Camera.UID	12606478859240980328	XmpString	Globally unique image identifier. Unsigned integer,

			must be a random number uniformly distributed in the range [0, 2^{64}-1]. The same random sequence must not be repeated in different units or for different acquisitions.

(XMP) Radiometric correction

Key	Value example	Type	Meaning
Xmp.Camera.BandName	{"Red", "NIR"}	XmpSeq	Name of each band (sequence). Single page image: one name for each band of the image. Multipage image: one band name per image page. (Note: this allows only one band per page since the sequence of the sequence is not possible). The band name must not start with the following characters: 0-9, ',', '(' or ')'.
Xmp.Camera.CentralWavelength	{660, 800}	XmpSeq	Central wavelength of each band in nm. Similar to band names. If band names are not defined, default band names are generated.
Xmp.Camera.WavelengthFWHM	{10, 10}	Xmp	Full-width half maximum of the

		Seq	wavelength distribution in nm.
Xmp.Camera.BlackCurrent	{10, 7}	XmpSeq	Dark current for each band/page: Camera radiometric correction specifications.
Xmp.Camera.BandSensitivity	{0.78, 0.043}	XmpSeq	The sensitivity of each band (pixel intensity value per irradiance).
Xmp.Camera.SunSensor	{12.87, 9.72}	XmpSeq	Sun irradiance sensor (upwards looking sensor) raw value for each band.
Xmp.Camera.SunSensorExposureTime	0.1	XmpText	Sun irradiance sensor exposure time in seconds.
Xmp.Camera.SunSensorSensitivity	{16.45, 14.08}	XmpSeq	Sun irradiance sensor sensitivity.
Xmp.Camera.InvalidPixel	{"40,593,981,291"}	XmpSeq	List of bad pixels as comma-separated x, y coordinate list, one sequence for each band.
Xmp.Camera.VignettingPolynomial	{"0.00325, 1.983e-6, 5.0983e-9"}	XmpSeq	Vignetting correction by radial polynomial: 1D polynomial coefficients (one string of comma-separated values per page, 1 string for RGB images): Camera radiometric correction specifications.
Xmp.Camera.VignettingCenter	{"542,912"}	XmpSeq	Vignetting correction by radial polynomial: Center point cx, cy in pixel (one string of comma-separated values per

			page): Camera radiometric correction specifications.
Xmp.Camera.VignettingPolynomial2DName	{"0,0,1,0, ..., 2,2,3,1"}	XmpSeq	Vignetting correction by 2D polynomial: name of coefficients: n,m for c_{n,m} separated by commas (one string of comma-separated values per page): Camera radiometric correction specifications.
Xmp.Camera.VignettingPolynomial2D	{"0.6332876761 49, 1.47936368033, ..., -1.58329242609, 0.130259802058"}	XmpSeq	Vignetting correction by 2D polynomial: coefficients c_{n,m} for names given by Xmp.Camera.VignettingPolynomial2DName separated by commas (one string of comma-separated values per page): Camera radiometric correction specifications.
Xmp.Camera.ColorTransform	{1, -0.114886,-0.0456918, -0.176716, 1, 0, -0.0715561, 0, 1}	XmpSeq	Sequence of double, color conversion matrix (for RGB: R2R, G2R, B2R, R2G, G2G, B2G, R2B, G2B, B2B). Considering the information as a 3x3 matrix M, $(R,G,B)_{new} = M * (R,G,B)$.
Xmp.Camera.IsNormalized	1	Boolean	Defines whether the radiometric correction is required (value 0) or the data has already been corrected (value 1) and therefore should not be corrected.

(XMP) Radiometric calibration target

Key	Value example	Type	Meaning
Xmp.Camera.Albedo	{0.22, 0.19, 0.37}	Xmp Seq	For radiometric calibration target images: Albedo of the disk for each band/page.
Xmp.Camera.ReflectArea	{"X1,Y1,X2,Y2,X3,Y3,X4,Y4", "X1,Y1,...", ...}	Xmp Seq	For radiometric calibration target images: polygon defining the target area for each page.
Xmp.Camera.CalibrationPicture	2	Short	Specifies if this image is a radiometric calibration target: 0 = no, 1 = yes but invalid, 2 = yes and valid. For Sequoia cameras: 0 = no, 1 and 2 = yes.

(XMP) Position and orientation

Key	**Value example**	**Type**	**Meaning**
Xmp.Camera.Yaw	72.7942	Rational or Double	Orientation data in degrees. For more information on the definition of yaw, pitch, roll: Yaw, Pitch, Roll and Omega, Phi, Kappa angles.

Xmp.Camera.Pitch	1.7943	Rational or Double	Orientation data in degrees. For more information on the definition of yaw, pitch, roll: Yaw, Pitch, Roll and Omega, Phi, Kappa angles.
Xmp.Camera.Roll	2.8909	Rational or Double	Orientation data in degrees. For more information on the definition of yaw, pitch, roll: Yaw, Pitch, Roll and Omega, Phi, Kappa angles.
Xmp.Camera.HorizCS	EPSG: 4326 EPSG: 6318	XmpString	The horizontal coordinate system used by the GPS receiver, i.e. by Exif.GPSInfo.GPSLatitude and Exif.GPSInfo.GPSLongitude, in the form auth:WKID, where auth is an authority (e.g. EPSG or ESRI) and WKID a well-known code.
Xmp.Camera.VertCS	EPSG: 5773 EPSG: 5703	XmpString	The vertical coordinate system used by the GPS receiver, i.e. by Exif.GPSInfo.GPSAltitude, in the form auth:WKID, where auth is an authority (e.g. EPSG or ESRI) and WKID a well-known code, or the special value ***ellipsoidal*** for ellipsoidal height.
Xmp.Camera.GPSXYAccuracy	5	Rational	Horizontal GPS Accuracy in meters.
Xmp.Camera.GPSZAccuracy	10	Rational	Vertical GPS Accuracy in meters.

Xmp.Camera.GyroRate	0.0487	Rational	Gyro rate during image acquisition, lower = better quality.
Xmp.Camera.IMUPitchAccuracy	0.113	Rational	The value in degrees of 1-sigma of the Gaussian noise of the IMU Pitch measurement.
Xmp.Camera.IMURollAccuracy	0.087	Rational	The value in degrees of 1-sigma of the Gaussian noise of the IMU Roll measurement.
Xmp.Camera.IMUYawAccuracy	1.250	Rational	The value in degrees of 1-sigma of the Gaussian noise of the IMU Yaw measurement.
Xmp.Camera.NominalCameraDistance	25.87	Rational	Nominal GPS distance between calibrated cameras in meters in order to have a decent overlap. Use to dynamically remove the cameras with worst quality as determined by the GyroRate such the target distance is as specified.
Xmp.Camera.AboveGroundAltitude	25.3	Rational or Double	Altitude above ground/take-off location in meters (from the altimeter of the drone).
Xmp.Camera.Location	[12.3759, -120.44, 12.294]	XmpSeq	Location as a 3-vector in the CRS given by *Xmp.Camera.CartesianCRS.*

Xmp.Camera.LocationAccuracy	[0.4395, 0.3442, 0.2938]	XmpSeq	Accuracy (sigma) of the location in the units given by the *Xmp.Camera.CartesianCRS*. If the CRS tag contains the special value *arbitrary*, the units are arbitrary but the same as for the location.
Xmp.Camera.Orientation	[-3.35472, 5.38485, 180.2475]	XmpSeq	Omega-Phi-Kappa angles in degree stating the orientation in the CRS given by *Xmp.Camera.CartesianCRS* . The convention is described in detail in this support page.
Xmp.Camera.OrientationAccuracy	[1.2394, 1.203, 1.2943]	XmpSeq	Accuracy (sigma) of the orientation Omega-Phi-Kappa angles in degree.
Xmp.Camera.CartesianCRS	EPSG:2227+5703	XmpString	CRS definition for the geolocation and orientation provided by the tags *XMP.Camera.Location* and *XMP.Camera.Orientation*. One of: • A WKT string version 2 describing a Cartesian 3D CRS. • A string in the format

			Authority:code+code where the first code is for a 2D CRS and the second one is for a vertical CRS (e.g. *EPSG:4326+5773*). • A string in the form *Authority:code+Authority:code* where the first code is for a 2D CRS and the second one is for a vertical CRS. • A string in the form *Authority:code* where the code is for 3D CRS. • The special value *arbitrary* for a completely arbitrary isometric 3D CRS without any scale.

(XMP) Sun irradiance position and orientation

Key	Value example	Type	Meaning
Xmp.Camera.SunSensorYaw	72.7942	Rational or Double	Sun irradiance sensor orientation data in degrees. For more information on the definition of yaw, pitch, roll: Yaw, Pitch, Roll and Omega, Phi, Kappa angles.
Xmp.Camera.SunSensorPitch	1.7943	Rational or Double	Sun irradiance sensor orientation data in degrees. For more information on the definition of yaw, pitch, roll: Yaw, Pitch, Roll and Omega, Phi, Kappa angles.
Xmp.Camera.SunSensorRoll	2.8909	Rational or Double	Sun irradiance sensor orientation data in degrees. For more information on the definition of yaw, pitch, roll: Yaw, Pitch, Roll and Omega, Phi, Kappa angles.
Xmp.Camera.SunSensorRelativeRotation	175, 0, 0	XmpSeq	Relative angle between camera and sunshine sensor in the omega, phi, kappa system (same as rig relatives).

(XMP) Bit-depth reduction

Key	Value example	Type	Meaning
Xmp.Camera.TransformAlpha	{55684.1, 58672.8, 58889.1}	XmpSeq	Sequence of double, alpha parameter for each band. This and the following two tags are used to undo a gamma transform which has been applied to fit data in 8bit space without

			losing the dynamic range: pixel_final = alpha * pixel_normalized^gamma + beta; where pixel_normalized is the pixel value stored in the JPG image scaled to the [0,1] interval.
Xmp.Camera.TransformBeta	{9850.9, 6862.2, 6645.91}	XmpSeq	Sequence of double, beta parameter for each band.
Xmp.Camera.TransformGamma	{2.12109, 2.6243, 2.33549}	XmpSeq	Sequence of double, gamma parameter for each band.

(XMP) Miscellaneous

Key	Value example	Type	Meaning
Xmp.Camera.SensorBitDepth	14	Short	Bit depth of the original sensor
Xmp.Camera.SensorTemperature	43.4	Double	Temperature of the sensor (degree Celsius)

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